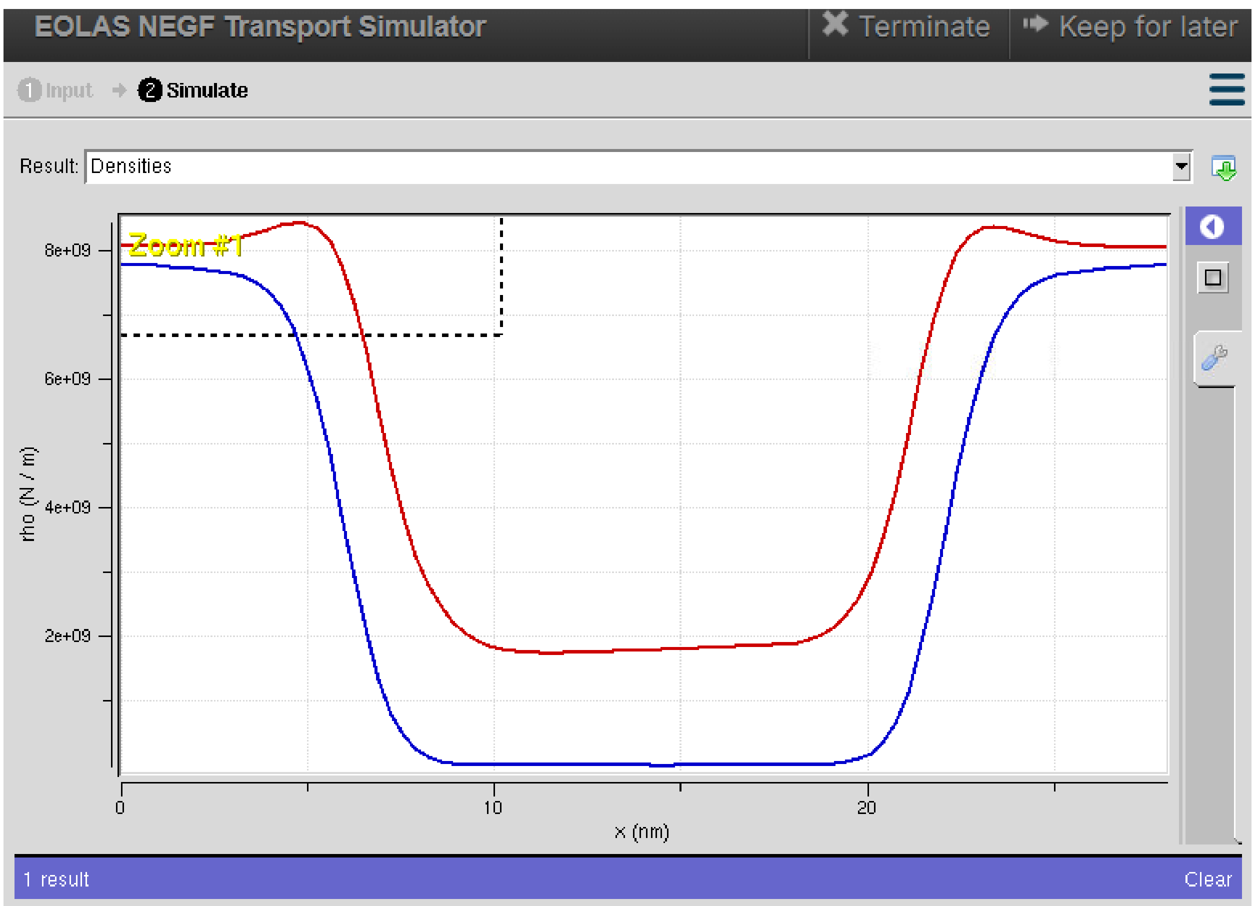
Exp. 8. Simulation of VI Characteristics of FINFET, CNTFET and GFET using NanoHub

(a) FINFET

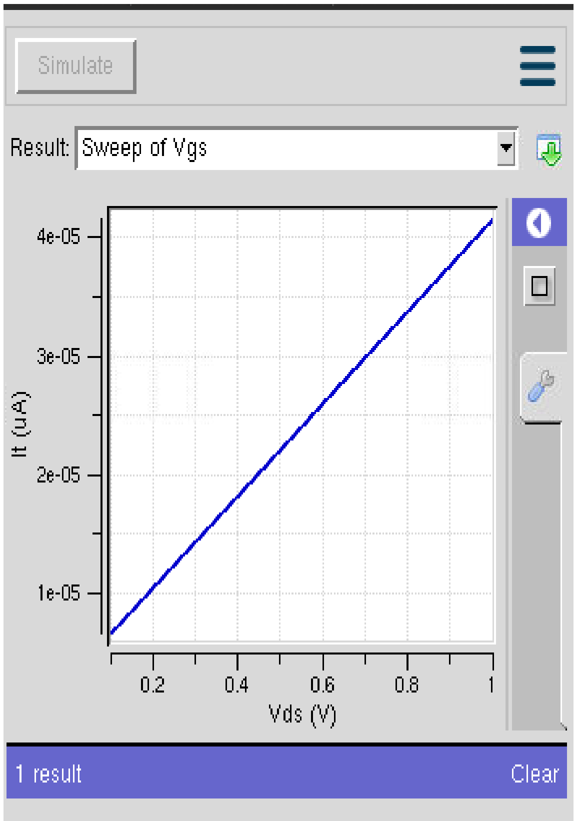
Case Study 1: Simulate & obtain the current spectrum characteristics for the last bias point of FinFET. Analyze the performance for different Oxide layer thicknesses

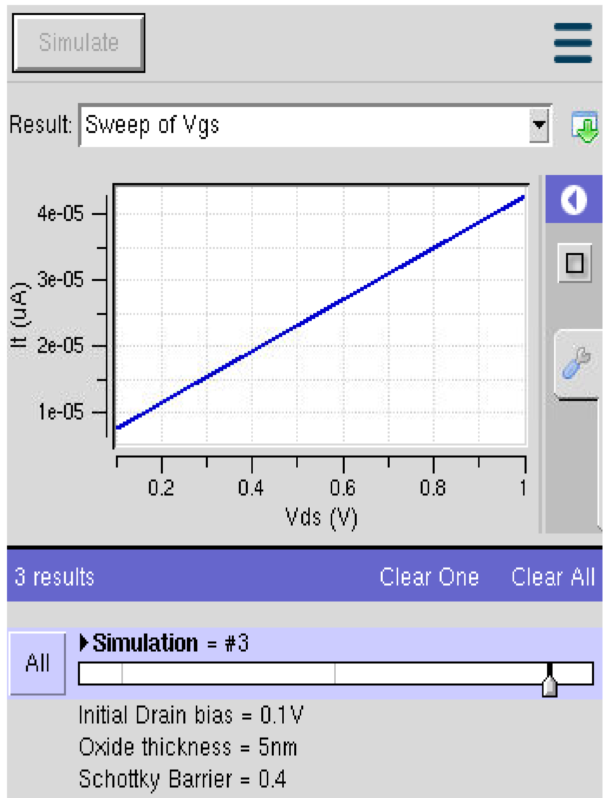
Case Study 2: Simulate & obtain the electron density per sheet characteristics of FinFET. Analyze its performance with different Gate region length



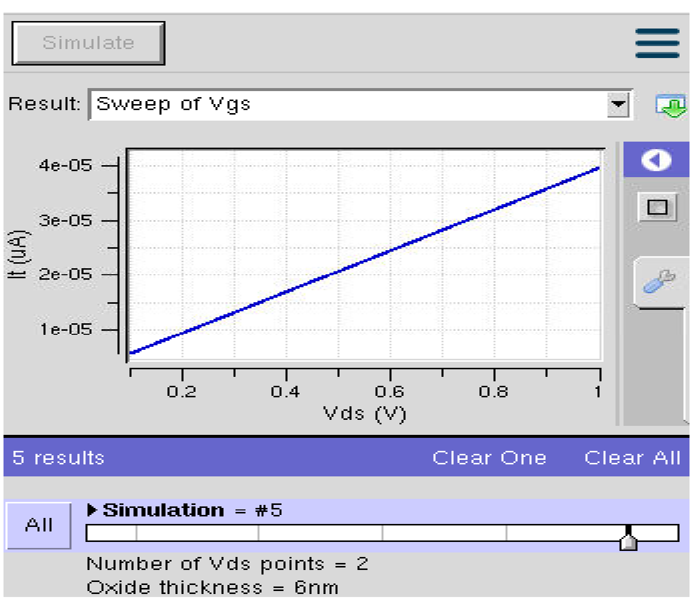
(b) CNTFET

Case study 1: Simulate & Obtain the VI characteristics of CNT-FET by varying Schottky barrier and oxide thickness



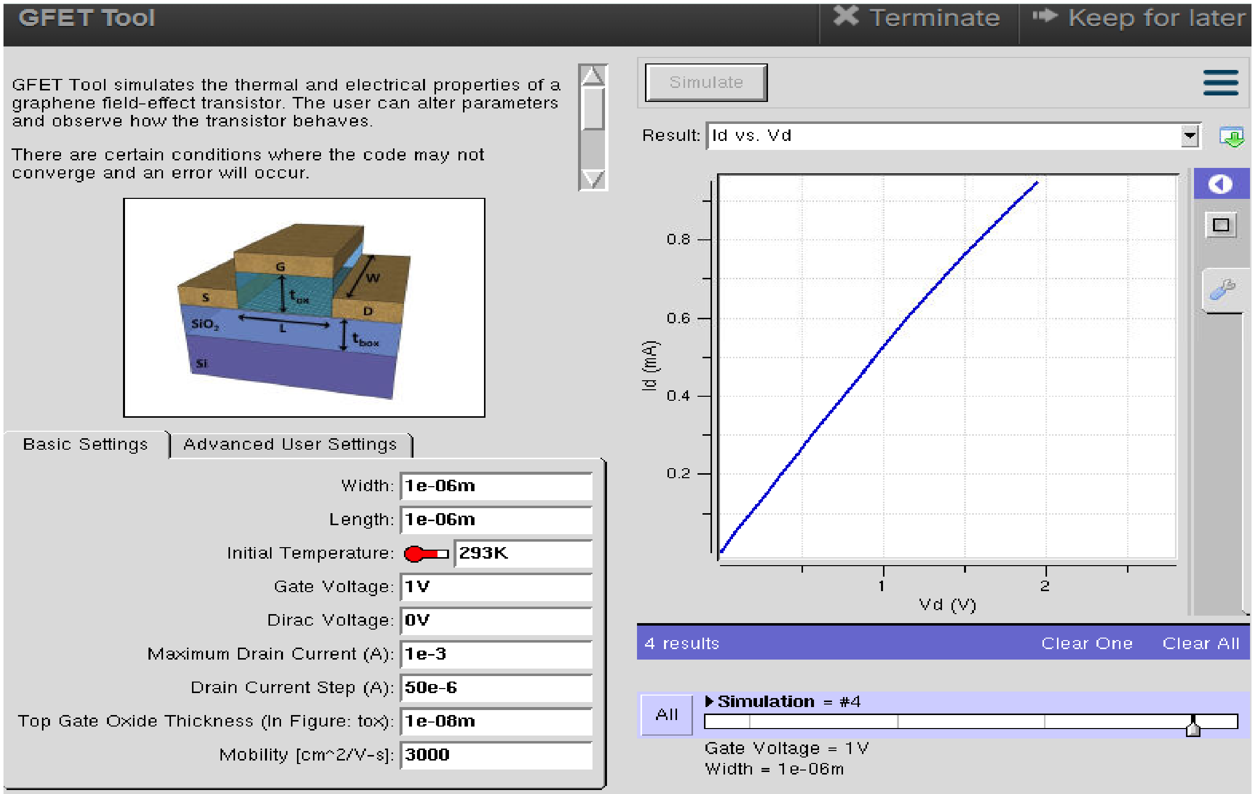


Case study 2: Simulate & Obtain the VI characteristics of CNT-FET by varying CNT diameter

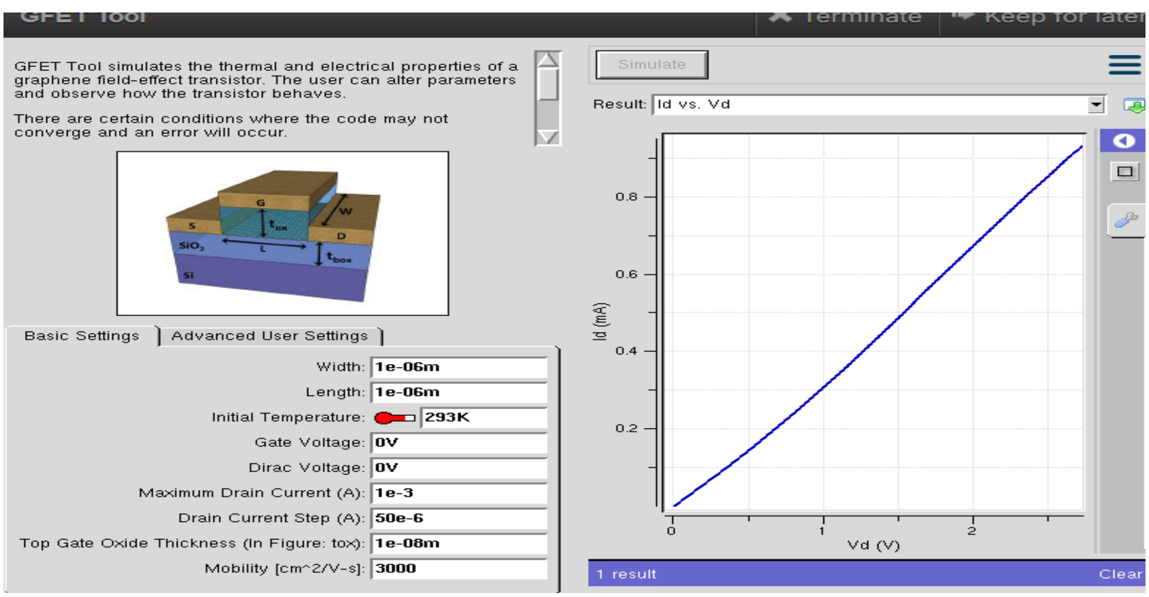


(c) GFET

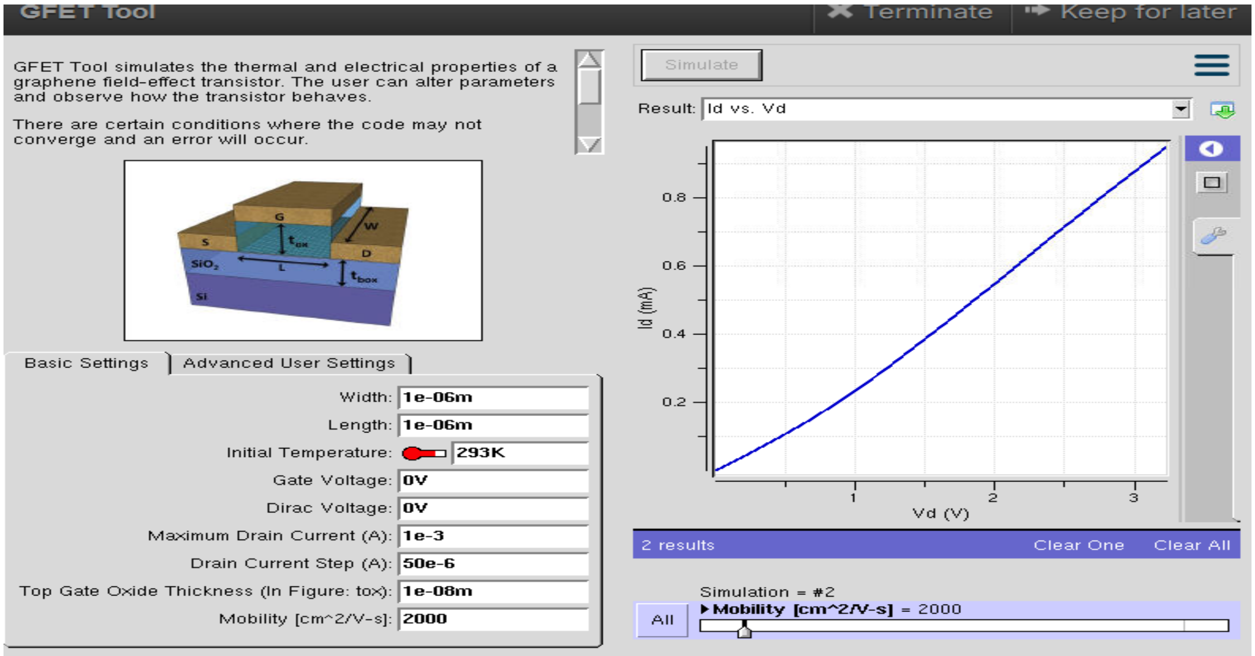
Case Study 1:How does varying the gate voltage (Vgs) affect the drain current (Ids) in a GFET?



Case Study 2: What factors determine the maximum drain current (Ids,max) in a Graphene Field Effect Transistor (GFET)?

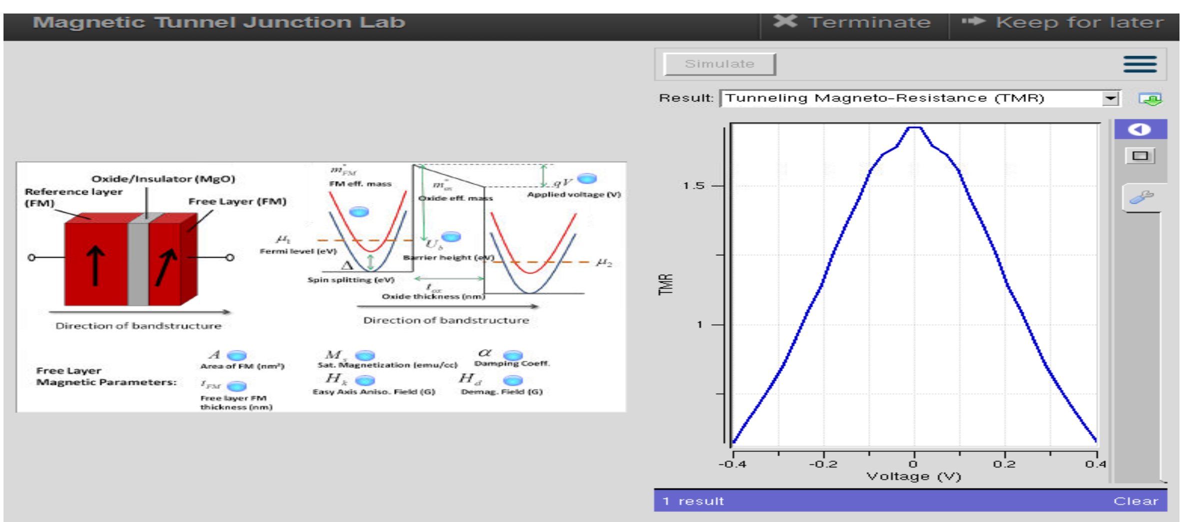


Case Study 3: To investigate how the mobility of charge carriers in a Graphene FieldEffect Transistor (GFET) with a mobility of 2000 cm²/V·s influences its current-voltage (VI) characteristics.

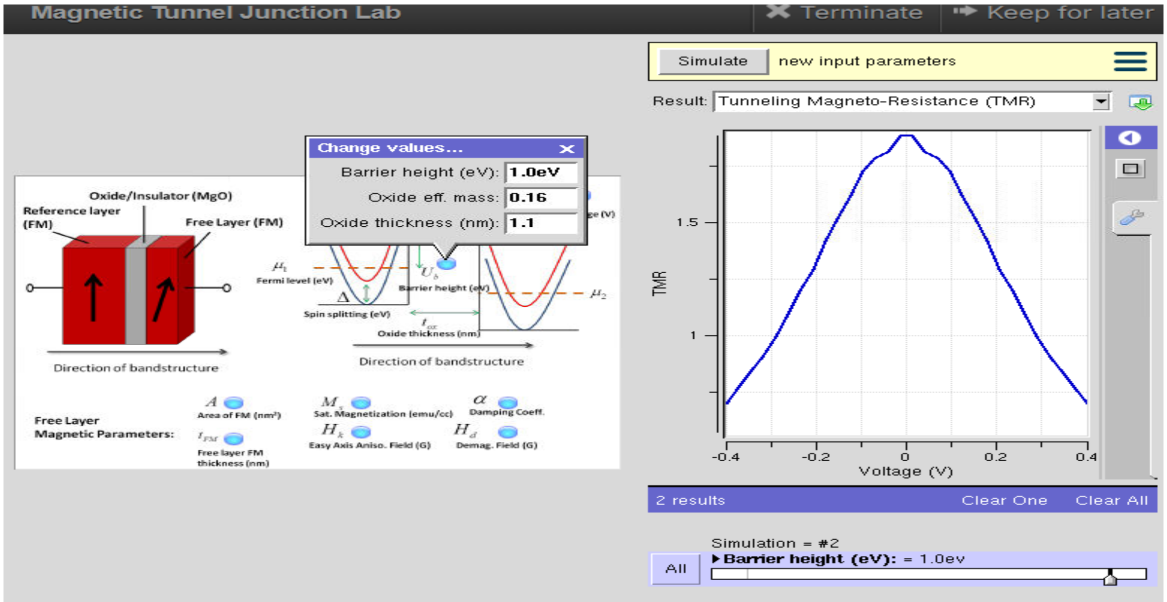


Exp. 9. Simulation of Tunneling Magnetoresistance (TMR) by Varying Barrier Height using NanoHUB

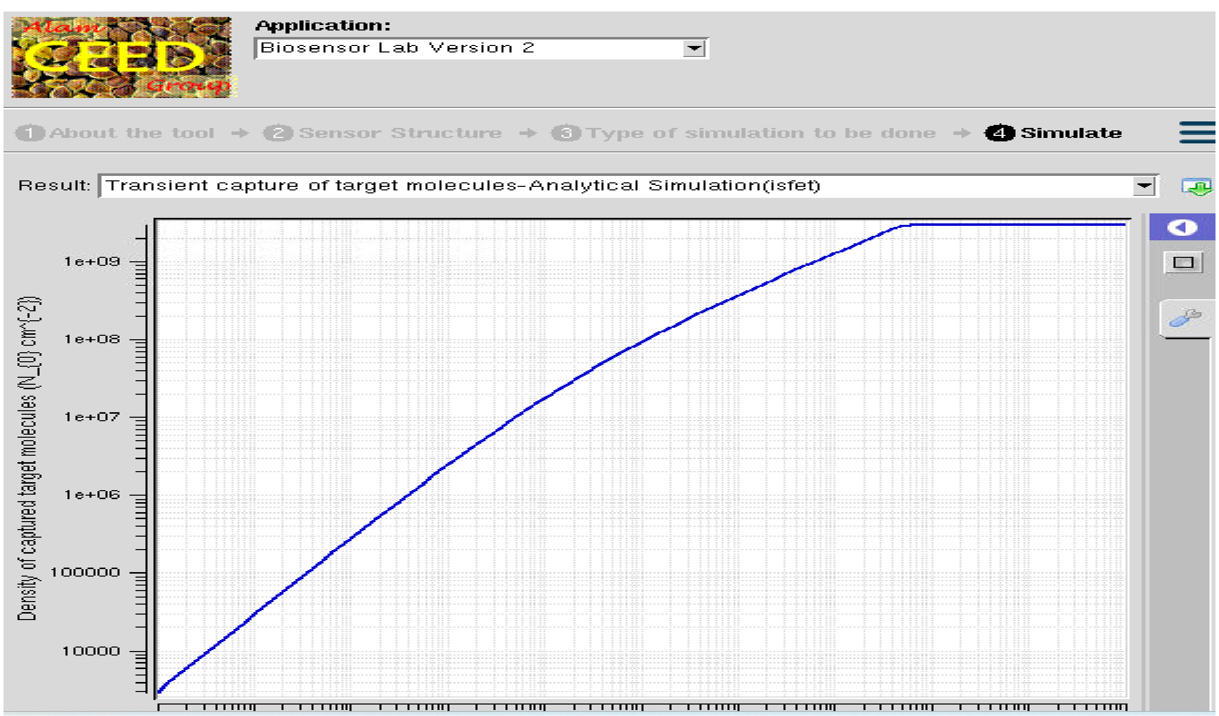
Case Study 1: Characteristics of TMR with barrier height of 0.85eV

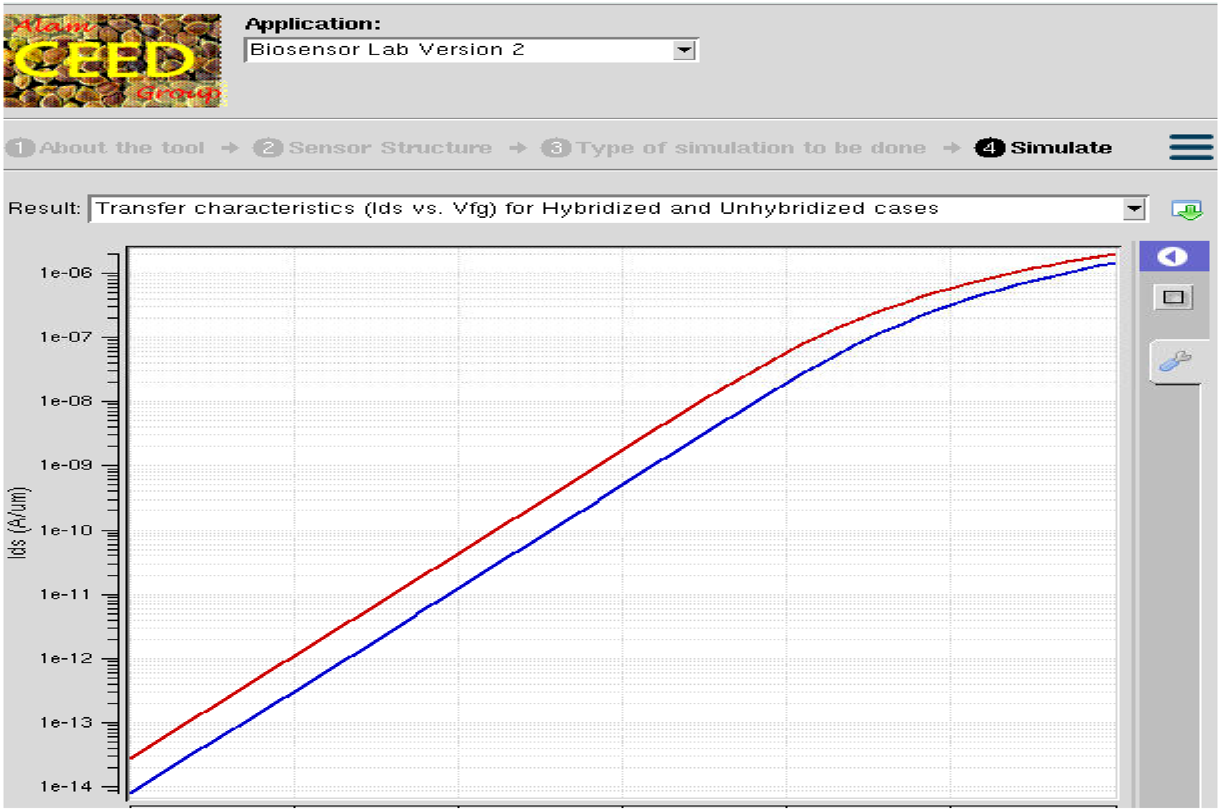


Case Study 2: Characteristics of TMR with barrier height of 1.0Ev



Exp. 10. Biosensor and Bio-FET characteristics using NanoHub




Application:
Biosensor Lab Version 2
1 About the tool
2 Sensor Structure
3 Type of simulation to be done
4 Simulate
Result: Transfer characteristics (Ids vs. Vfg) for Hybridized and Unhybridized cases
Ids (A/um)
1e-06
1e-07
1e-08
1e-09
1e-10
1e-11
1e-12
1e-13
1e-14